ResCap: Fast-yet-Accurate Capacitance Extraction for Standard Cell Design by Physics-Guided Machine Learning

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ABSTRACT

In the field of VLSI design, accurate capacitance extraction is essential for ensuring optimal performance of integrated circuits, especially in standard cell designs. Conventional techniques, such as the 2.5D model and 3D field solver, either suffer from inaccuracies or are computationally intensive. To address these challenges, we present ResCap, an innovative approach that synergizes physics-guided linear models with advanced machine learning techniques. Rather than directly predicting the target capacitance, our method starts by employing physical principles to estimate the initial capacitance value, ensuring that predictions are grounded in well-established physical laws. Subsequently, machine learning is applied to predict residual values, thereby refining the initial estimates. This approach not only enhances accuracy and generalization but also reduces dependency on extensive training datasets. Experimental results demonstrate that ResCap significantly outperforms conventional methods on industrial standard cell designs under 4nm process technology, achieving high accuracy with an average error of 0.06% in delay and 0.16% in power. Notably, ResCap exhibits no outliers (error > 1%), whereas the conventional 2.5D extraction tool shows significant outliers of 10.95% in delay and 50.4% in power. Furthermore, our framework demonstrates remarkable efficiency, reducing extraction time by 215x compared to field solvers.

KEYWORDS

Parasitic capacitance extraction, Physics-guided machine learning, Deep neural-networks, Standard cell PPA

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1 INTRODUCTION

In VLSI design, standard cells are fundamental components that significantly influence the power, performance, and area (PPA) of integrated circuits. As technology nodes shrink, optimizing these standard cells becomes essential to meet product requirements. Interconnect capacitance plays a crucial role in the optimization of standard cells, as higher layout density in advanced nodes results in more parasitic elements. Consequently, the effects of these parasitics are no longer negligible, necessitating highly accurate parasitic extraction to prevent timing errors, excessive power consumption, and signal integrity issues [18].

Conventional parasitic extraction methods include 2.5D and 3D approaches. The 2.5D method uses pattern matching and model fitting based on pre-characterized libraries, offering high performance but lower accuracy. This becomes crucial due to the PPA of standard cells in advanced nodes is highly sensitive to capacitance [13]. As depicted in Fig. 1, variations in capacitance ranging from 1% to 10% lead to substantial fluctuations in both delay and power, with delay varying from 0.3% to 3.7% and power from 0.5% to 6.3%. Typically, standard cell designs can only tolerate a 1% difference in both delay and power, highlighting the necessity for more precise extraction methodologies. To achieve higher accuracy, the 3D field solver is utilized to model the complex geometries and interactions within standard cells. The solver calculates detailed capacitance

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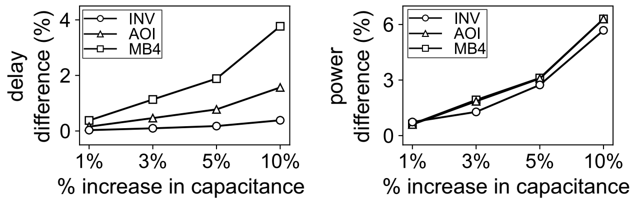

Figure 1: Impact of capacitance variations on standard cell delay and power

values by solving physical equations that describe electric and magnetic field behaviors. However, the computational complexity of the 3D field solver poses challenges, particularly during the design optimization phase, where iterative refinement and extraction are necessary to fulfill design requirements.

Various machine learning techniques have been proposed to address the challenge of predicting capacitance values directly from layout patterns. These methods leverage extensive datasets of precharacterized layouts to train models capable of efficiently predicting capacitance for new designs. The work [7] introduced a neural network-based approach for 3D IC structures using MLP networks, focusing on single-dielectric structures but with some errors exceeding 10% for total capacitance. CNN-Cap [16, 17] employed a convolutional neural network with a ResNet architecture, offering precise capacitance predictions for 2D and 3D structures, surpassing traditional methods in precision and simplicity. Deep Neural Network (DNN) models have been developed for 2D patterns, focusing on tasks such as pattern classification and formula building [9], metal connectivity [3], process variations [5] and matrix-based coupling capacitances [11], improving the accuracy and efficiency of traditional 2.5D rule-based methods. A hybrid approach [4] combining field-solver, rule-based, and 3D pattern-based DNN models has been proposed to meet user accuracy requirements by identifying the optimal extraction method. GNN-Cap [10] employed a graph neural network to model intricate 3D interconnect structures as graphs, capturing spatial and geometric information for precise and efficient capacitance prediction.

Although data-driven machine learning approaches offer significant speed advantages, they face challenges related to the quality and diversity of training data. Models trained on limited or biased datasets may struggle to generalize to new or complex layouts, resulting in inaccuracies in predicting capacitance. Additionally, these models often function as black boxes, providing limited insight into the underlying physical phenomena. To address these limitations, a novel hybrid approach, ResCap, is proposed in this study, which combines physical laws with machine learning techniques. Instead of directly learning layout patterns for capacitance prediction, the proposed method initially calculates capacitance using a physics-guided linear model. Subsequently, machine learning is employed to predict *residual* values, thereby enhancing the initial estimates. This innovative concept is depicted in Fig. 2.

Figure 2 (a) shows the histogram of one of the dominant capacitances (same-layer lateral capacitance), which has been utilized directly in previous machine learning work. By examining the overlap

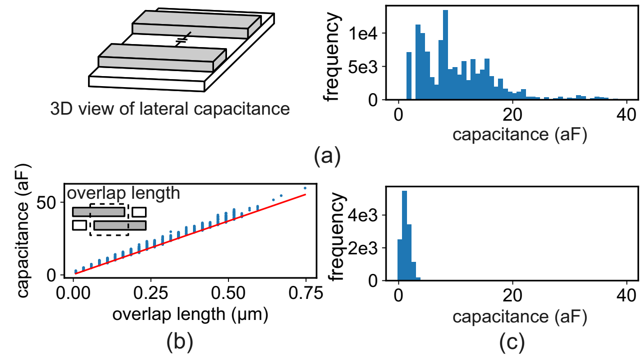

Figure 2: (a) The histogram of same-layer lateral capacitances used in prior machine learning work. (b) Linear relationship between overlap length and capacitance. (c) The histogram of residual capacitances derived from the linear model.

length of layout patterns, as depicted in Fig. 2 (b), a linear relationship between the overlap length and capacitance is revealed. This insight motivates us to adopt a straightforward, physics-grounded approach to identify an appropriate reference linear equation for initial estimation, subsequently enhancing its accuracy by utilizing a learning model to predict the residual value. In this study, the linear equation is derived from a 3D field solver applied to a *3D well-mesh structure*, as indicated by the red line in Fig. 2 (b). As a result, the model's focus shifts from predicting the original capacitance to concentrating on the *residual* capacitance, as shown in Fig. 2 (c). This physics-guided strategy brings multiple benefits, such as increased accuracy, better generalization, and less reliance on extensive training datasets. In conclusion, our approach not only mitigates the limitations of purely data-driven models but also provides deeper insights into the underlying physical phenomena. The main contributions of this work can be summarized as:

(1) Integration of Physics-Guided and Machine Learning Models: ResCap combines the strengths of physics-based models and machine learning, providing a reliable initial estimation grounded in physical laws and refining it with machine learning to achieve high accuracy.
(2) Consideration of Via Effects: ResCap incorporates the impact of vias during capacitance extraction, significantly reducing the error of both coupling (4.5% to 1.85%) and total capacitance (2.89% to 0.77%) predictions, which is crucial for advanced technology nodes.
(3) Fast-Yet-Accurate Capacitance Extraction: ResCap demonstrates superior performance in achieving errors in delay and power to only 0.06% and 0.16%, respectively, while also achieving a 215x reduction in extraction time compared to conventional 3D field solvers, making it highly practical for iterative design processes.

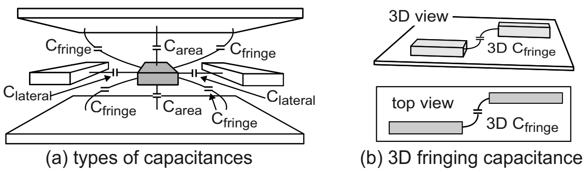

Figure 3: (a) An illustration of parasitic capacitances, including area, lateral, and fringing capacitances. (b) An example of 3D fringing capacitances that are not captured in 2D pattern extraction methods.

2 BACKGROUND

2.1 2.5D Model and 3D Field Solver for Parasitic Capacitance Extraction

The 2.5D capacitance extraction method is a widely utilized approach for balancing accuracy with computational efficiency. This method involves sweeping 3D interconnect geometries across 2D planes (e.g., z-x and z-y planes) to capture essential capacitance components such as area, lateral, and fringing capacitances, as depicted in Fig. 3 (a). It consists of two primary steps: characterization and extraction. During characterization, interconnect patterns are generated based on process technology, and their capacitances are calculated using detailed 3D field solvers, with results stored in a pre-characterized library. In the extraction phase, the layout is scanned, and capacitances are determined by matching layout patterns to library entries.

Despite its computational efficiency, the 2.5D model has some limitations. It relies on a pre-characterized library that may not cover all layout patterns, leading to potential inaccuracies in complex designs. Additionally, its dependence on simplified 2D patterns may miss critical 3D interactions, failing to accurately reflect the true electrical behavior of advanced 3D geometries, such as 3D fringing capacitance, as illustrated in Fig. 3 (b).

The 3D field solver approach provides a detailed and accurate solution to model the electromagnetic coupling effects by solving numerical equations that satisfies the following equation.

$$C = Q/V \tag{1}$$

where C is capacitance, Q is charge, and V is voltage. The methodologies employed in 3D field solvers encompass the finite difference method (FDM) [15], the boundary element method (BEM) [12], and the floating random walk (FRW) [8, 19]. The 3D field solvers excel in capturing complete 3D geometry and material properties, effectively managing new and complex layouts. However, they demand substantial computational resources, including processing and memory.

2.2 Parasitic Capacitance Extraction for Standard Cell in Advanced Nodes

In advanced nodes, process shrinkage reduces area and tightens spacing between layout patterns, thereby increasing the significance of coupling capacitance. This effect is particularly pronounced in standard cell designs, where the emphasis on minimizing area results in a higher layout density, making certain factors significant in accurate parasitic capacitance extraction. Firstly, the impact of surrounding metals and vias on parasitic capacitances is significantly heightened due to the increased density and reduced dimensions of interconnects. Moreover, the middle-end-of-line (MEOL) capacitance assumes greater importance as the back-end-of-line (BEOL) interconnects in standard cell become shorter [6, 18]. Furthermore, mask misalignments from double patterning introduce additional geometric variations that affect capacitances, necessitating their inclusion in the analysis. To address the aforementioned challenges, a 3D pattern structure that encompasses the MEOL, BEOL, and via layers is considered in this work. The double patterning effect is managed by isolating each mask into different layers.

However, smaller geometries pose challenges but also offer opportunities due to tighter design rules, resulting in more regular and directional layout patterns aligned only in the x or y direction, while eliminating L-shaped patterns. As a result, this regularity simplifies the integration of physical rules with machine learning algorithms, enabling more accurate capacitance extraction.

3 RESCAP MODELLING FRAMEWORK

The ResCap modelling flow for capacitance extraction based on machine learning is systematically illustrated in Fig. 4. This process consists of three main phases: (1) developing a pre-characterized physics-guided linear model to establish an accurate initial estimation; (2) preparing training data, including layout pattern extraction, data augmentation, processing to incorporate the pre-characterized physics-guided model, and conversion to a 3D irregular-grid-based layout representation; and (3) training the model using DNN.

3.1 Physics-Guided Linear Model

Our approach begins with the formulation of a pre-characterized, physics-guided linear model to estimate the initial capacitance values for each overlap layout pattern. This model leverages fundamental physical principles to provide a baseline capacitance estimation, ensuring that the predictions are grounded in well-established physical laws. Equation 1 can be expressed as the common parallel plate capacitance formula,

$$C = \epsilon \cdot w \cdot L/d = k \cdot L \tag{2}$$

In this equation, the capacitance C is exactly proportional to the overlap length L between two parallel conductors. The proportionality constant k could be derived by fixing dielectric ϵ, width w, and distance d. In standard cell layouts, w and d are easy to be fixed by modeled as different scenarios. The challenge lies in fixing ϵ for the linear model construction. To address this, we propose a 3D well-mesh structure for the target victim and aggressor, as illustrated in Fig. 5. This structure features a densely packed arrangement of metal wires in a 3D layout, where all surrounding metals —including top, bottom, and neighboring metals of the target polygons—are fully filled while adhering to minimum spacing rules, as shown in Fig. 5 (a). Moreover, any metal layer between the targets is removed to avoid the shielding effect, as shown in Fig. 5 (b). Note that the well-mesh structure is used due to its proximity to actual layout design, as standard cell typically exhibit a high-density layout.

Based on the well-mesh structure, we adjust only the overlap length between the victim and aggressor conductors and run the 3D

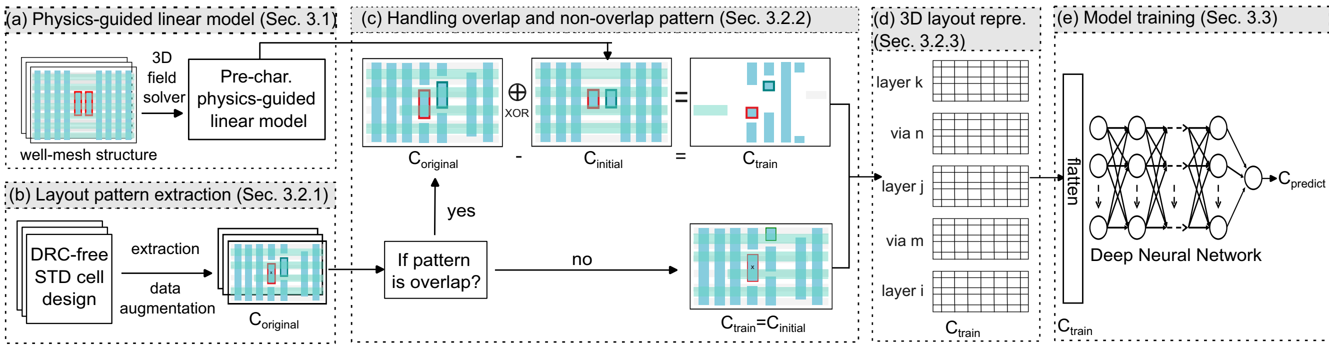

Figure 4: ResCap modelling framework consists of (a) physics-guided linear model (b) layout pattern extraction and data augmentation (c) handling overlap and non-overlap pattern (d) 3D layout representation and (e) model training

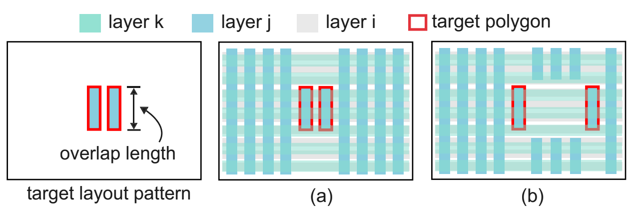

Figure 5: An example to illustrate 3D well-mesh structure: (a) all surrounding metals of the target polygons, including top, bottom, and neighbor, are fully filled (b) removal of the metal between target polygons to avoid the shielding effect.

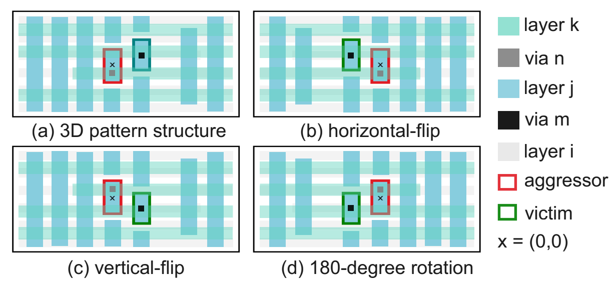

Figure 6: Data augmentation of the original 3D pattern structure: (a) original structure, (b) horizontal flip, (c) vertical flip, (d) 180-degree rotation.

field solver to obtain the capacitance value. The capacitance value adheres to physical laws and forms a linear model, requiring only a few data points (e.g., 5 in our work) for model construction. In summary, the physics-guided linear model serves as the foundation for our capacitance extraction process. It provides a reliable initial estimation that is both computationally efficient and physically accurate, setting the stage for subsequent machine learning-based refinement.

3.2 Data Processing and Representation

The preparation of training data is a critical step in developing an accurate and robust machine learning model for capacitance extraction. This phase involves three main tasks: layout pattern extraction, handling of overlap and non-overlap layout patterns, and 3D layout representation.

3.2.1 Layout Pattern Extraction and Augmentation. In this study, to achieve more realistic layout patterns, the training dataset is extracted from hundreds of real standard cell designs. Capacitance values for each element within these designs are simulated using 3D field solver. To account for additional effects in advanced nodes (ex. 3D fringing capacitance and via effect), a 3D pattern structure is considered, incorporating three metal layers and two via layers. An example of top view of this 3D pattern is depicted in Fig. 6 (a) where the aggressor is positioned at the center of the box. During the extraction process, the layout of the real design is systematically scanned layer by layer and polygon by polygon. When a polygon is selected as the aggressor, a capture box is employed to encapsulate the 3D pattern, with the aggressor placed at the center while other polygons within the box are designated as victims. The coupling capacitance between each aggressor and victim pair is then determined based on the capacitance values obtained from the 3D field solver previously. It is important to note that the size of the capture box is process-dependent and could be determined by simulating the spacing between neighboring conductors until the coupling capacitance is less than 1% of that simulated from minimum space.

To further enhance the robustness and generalization of our learning model, data augmentation techniques are employed. Specifically, we flip and rotate the layout pattern to generate additional training samples without any loss of accuracy, as capacitance depends solely on the surrounding environment rather than specific polygon coordinates. As illustrated in Fig. 6 (b)-(d), three new training samples are created by 'horizontal-flip', 'vertical-flip', and '180-degree rotation', while maintaining consistent coupling capacitance between the aggressor and target victim. This augmentation technique effectively quadruples the training samples from the original dataset, enhancing the model's performance and generalization.

3.2.2 Handling the overlap and non-overlap pattern. Each aggressor and victim pair in the layout pattern can be categorized into two distinct types: overlap and non-overlap patterns. The overlap

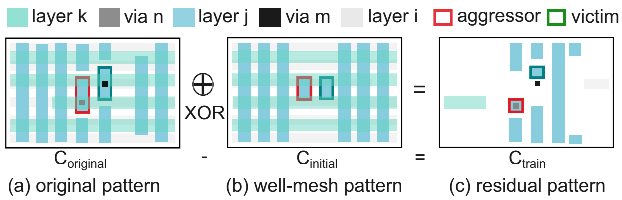

Figure 7: An example demonstrating the XOR operation for generating residual layout patterns used in training.

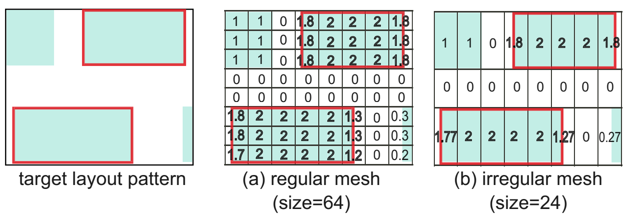

Figure 8: 3D grid-based layout representation (a) regular mesh (b) proposed irregular mesh

pattern in this work is defined as a scenario where two polygons overlap if and only if their projections onto the three principle axes (x, y, and z) overlap. The overlap patterns, often referred to as area and lateral capacitance, typically dominate the capacitance values, while non-overlap patterns, known as fringing capacitance, contribute less significantly.

Overlap Pattern For overlap patterns, we could leverage our pre-characterized physics-guided linear model to calculate the initial capacitance, $C_{initial}$. Given the accurate initial estimate of the capacitance value, our training focus is on the residual capacitance rather than the original capacitance, $C_{original}$. This can be expressed as:

$$C_{train} = C_{original} - C_{initial} \tag{3}$$

In addition to modifying the target from overall capacitance to residual capacitance, we also transform the layout pattern from the original to the residual one using an XOR operation with a well-mesh structure employed in the linear model. Fig. 7 demonstrates the example of deriving the residual layout pattern for model training, with subfigures (a) and (b) representing the original pattern and well-mesh pattern, respectively. By utilizing the XOR operation, the residual pattern (Fig. 7 (c)) is obtained as the discrepancy between the original and well-mesh patterns. This approach ensures that the machine learning model focuses on the residuals, thereby enhancing prediction accuracy and efficiency.

Non-overlap Patterns For non-overlap patterns, the capacitance values are generally lower. In this case, we directly apply the machine learning model to predict the capacitance without the need for an initial calculation. This streamlined approach simplifies the process and reduces computational overhead while maintaining accuracy. The training capacitance remains the same as the original coupling capacitance, as shown below,

$$C_{train} = C_{original} \tag{4}$$

Table 1: Hyperparameters search space

Hyperparameter	Range
Activation function	ReLu, Tanh
Number of hidden layers	1, 2, 3, 4, 5
Neurons in each hidden layers	range(16,512,16)
Learning rate	0.01, 0.001, 0.0001

3.2.3 3D Irregular-grid-based Layout Representation. Similar to previous work [4, 16], a grid-based (density-based) representation is used to simplify layout features while preserving geometric and spatial information, serving as the input for the relevant learning model. The value in each grid denotes the density, defined as the proportion of the grid occupied by the conductor. To precisely address aggressor and victim conductors, the corresponding grid will be incremented by one to identify their presence. A notable departure from previous approaches is the use of an irregular grid structure for layout representation. Earlier methods typically employ a regular mesh with a size defined as half of the minimum spacing of the metal layer, resulting in a finer grid and subsequently increasing the size of feature vectors (Fig. 8 (a)). This leads to escalated computational and memory requirements, potentially hindering the training speed and accuracy of machine learning models. In scenarios such as standard cell design in advanced nodes, where layout patterns are typically regular, the grid size can be determined based on process-dependent parameters like the minimum width and minimum spacing of the metal layer. An instance of irregular-grid-based layout representation is depicted in Fig. 8 (b), where the irregular mesh successfully reduces the grid size from 64 to 24. In summary, the adoption of an irregular mesh offers the advantage of achieving a reduced feature vector length while preserving the complete geometric information of layout patterns. Another difference from previous works is the comprehensive nature of our layout representation, including not only the top, bottom, and adjacent layers but also vias, which are crucial in advanced nodes.

3.3 Model Training

For the predictive modelling, we employ a DNN to capture the intricate relationships between the layout features and the parasitic capacitances. The inputs of the DNNs are the flattened irregular-grid-based layout representation of each metal and via layer of a layout pattern as shown in Fig. 4 (d) and (e). The output of the model is the coupling capacitance between two specified polygons within a given layout pattern. To accurately account for the multi-dielectric environment, we will develop different models trained on each combination of metal layers in the target process node. For each combination of metal layers, two models are created: one for overlap patterns and the other for non-overlap patterns. The DNN architecture and model parameters are chosen based on the Bayesian optimization (BO) algorithm [14] which has proven to be effective in optimizing DNN by adeptly managing complex and high-dimensional hyperparameter spaces.

4 EXPERIMENTAL RESULTS

The experiments in this work were conducted under 4nm process technology and incorporated numerous industrial standard cell designs. The golden capacitance was derived from the 3D field solver,

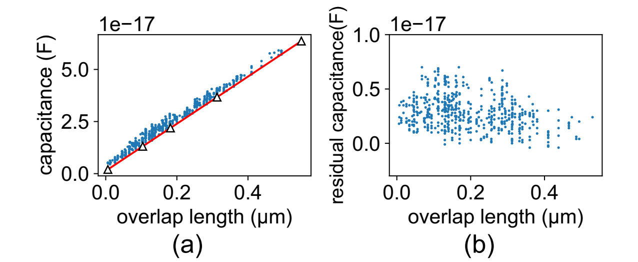

Figure 9: (a) Scatter plot of capacitance values versus overlap length, where the red line represents a physics-guided linear model generated from a limited dataset of 5 points (indicated by triangles). (b) Scatter plot of residual capacitance values versus overlap length, illustrating the linear model's effectiveness in decoupling the effect of overlap length.

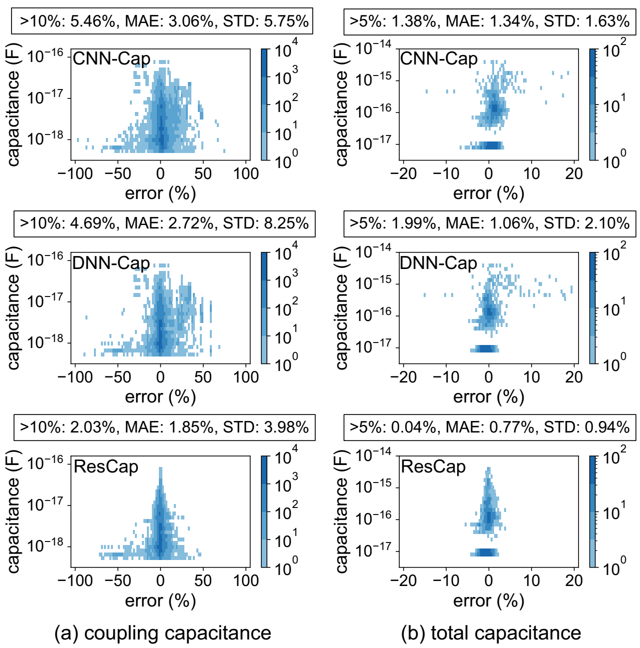

Figure 10: Heatmap of the relative error for (a) coupling capacitance and (b) total capacitance using CNN-Cap, DNN-Cap, and ResCap.

QRCFS [1]. Three approaches were implemented for benchmarking, namely DNN-Cap [4], CNN-Cap [16], and our proposed method, ResCap. Both DNN-Cap and ResCap were developed using TensorFlow on Intel Xeon Gold 6344, utilizing Bayesian optimization for DNN model architecture and parameter optimization. The search space for hyperparameters is outlined in Table 1. The CNN-Cap was implemented using PyTorch, based on [16], and executed on 8 Nvidia RTX2080Ti GPUs. Two ResNets, ResNet-34 and ResNet-18, were tested, with ResNet-18 performing slightly better in our test experiments. All models were trained on a consistent dataset comprising 800 industrial standard cell designs with 1.6 million capacitance components and evaluated on another 50 unseen cells.

4.1 Analysis of Physics-Guided Linear Model

Figure 9 (a) shows the scatter plot of the capacitance values and their associated overlap lengths of a primary metal layer. The red line represents a physics-guided linear model generated from a limited dataset of 5 points (indicated by triangles), which were simulated using a 3D field solver on the proposed well-mesh structure. As depicted in the figure, the linear model closely aligns with the actual capacitance values, emphasizing the dependability of these models as baseline references for further accurate capacitance predictions. Moreover, the relationship between residual capacitance values and the overlap length is also illustrated in Fig. 9 (b). The results suggest that there is no longer a correlation between the residual capacitance and the overlap length, indicating that the linear model has successfully decoupled the effect of overlap length. Consequently, this allows the DNN to concentrate more efficiently on learning the impact caused by the surrounding environment.

4.2 Extraction Result and Benchmark

We then benchmarked our approach, ResCap, against the state-of-the-art CNN-Cap and DNN-Cap approaches. Figure 10 presents the heatmap of the relative error for both (a) coupling capacitance and (b) total capacitance, along with the mean absolute error (MAE) and standard deviation of the relative errors (STD). Similar to prior studies [4, 16], the outlier criteria have been set to 10% error for coupling capacitance and 5% error for total capacitance. The experimental results indicate that ResCap significantly outperforms both CNN-Cap and DNN-Cap in estimating coupling capacitance and total capacitance. For coupling capacitance, ResCap achieved the lowest MAE of 1.85%, compared to 3.06% for CNN-Cap and 2.72% for DNN-Cap. Similarly, for total capacitance, ResCap exhibited superior performance with an MAE of 0.77%, while CNN-Cap and DNN-Cap had higher MAEs of 1.34% and 1.06%, respectively. Notably, the STD and outlier percentages of ResCap are significantly lower than those of CNN-Cap and DNN-Cap. Specifically, most outliers in coupling capacitance in ResCap are less than 1 aF, whereas outliers in DNN-Cap and CNN-Cap range from 1 aF to 100 aF. The reason is that high capacitance interconnects are mainly affected by lateral capacitance, which is accurately represented by our physics-guided linear model. This model offers a good initial estimate and specifically targets the prediction of residual capacitance within a narrower range compared to other methods, ultimately yielding better results. In summary, ResCap not only delivers more accurate capacitance predictions but also showcases enhanced reliability and consistency in its estimations.

The impact of vias on capacitance extraction was also investigated, as shown in Fig. 11. The results indicate that the incorporation of vias has a notable positive impact on the accuracy of both coupling and total capacitance extraction. Specifically, the inclusion of vias results in a decrease in the MAE and STD, as well as a reduction in the occurrence of outliers. For coupling capacitance, the MAE is reduced from 4.50% to 1.85%, with outliers decreasing from 8.95% to 2.03%. Similarly, for total capacitance, the MAE decreases from 2.89% to 0.77%, and outliers are completely eliminated. These findings demonstrate that vias cannot be excluded during capacitance extraction in standard cell design in advanced nodes.

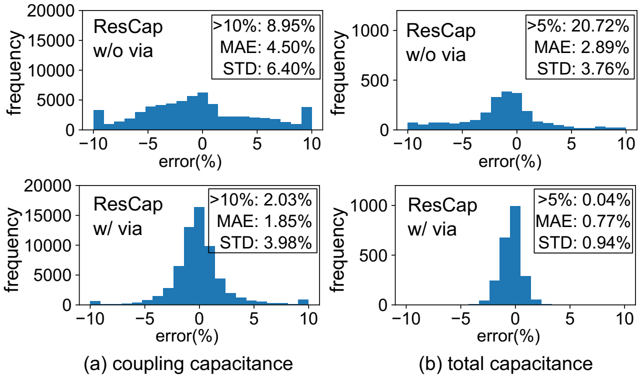

Figure 11: Impact of via on parasitic capacitance extraction (a) coupling capacitance (b) total capacitance.

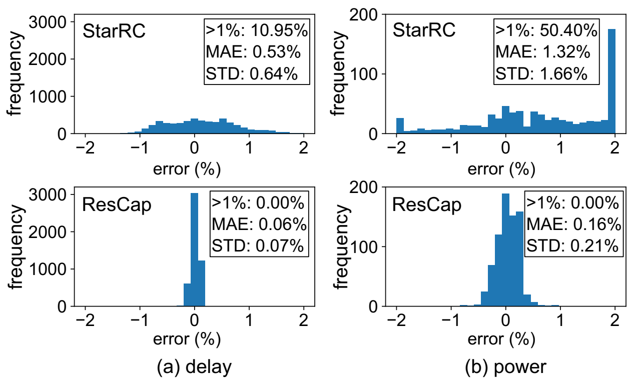

Figure 12: Comparison of delay and power at cell level with StarRC

4.3 Cell Level Analysis

To further validate the accuracy and efficiency of ResCap, a thorough examination was carried out to evaluate the delay and power error caused by the capacitance extracted from ResCap at the cell level, as shown in Fig. 12. This evaluation involved a comparative analysis with the 2.5D extraction tool StarRC [2], which is commonly utilized in the industry for standard cell extraction. The SPICE simulations were conducted on the various netlists extracted from ResCap, StarRC, and QRCFS to derive the delay and power values for each cell, with the values obtained by using QRCFS netlist serving as the benchmark for comparison. Experimental results demonstrate that ResCap significantly outperforms StarRC on 50 industrial standard cell designs, achieving high accuracy with an average error of 0.06% in delay and 0.16% in power. Notably, ResCap exhibits no outliers (error>1%), whereas the StarRC shows significant outliers of 10.95% in delay and 50.4% in power. It may result from pattern mismatches and the omission of certain 3D fringing capacitances in the 2.5D extraction process. Moreover, a runtime comparison between StarRC, ResCap, and QRCFS is listed in Table 2. ResCap demonstrates comparable runtime to StarRC and exhibits remarkable efficiency, reducing extraction time by 215X compared to QRCFS.

Table 2: Runtime between StarRC, QRCFS, and ResCap

Cell Group	StarRC (s)	ResCap (s)	QRCFS (s)	Speedup (x)
AND/OR	52	13	2313	177.9
AOI/OAI	57	17	3163	186.1
FA/HA	58	21	2975	141.6
INV/BUF	49	13	4299	330.7
LH	48	16	4420	276.3
MB2	54	41	5919	144.4
MB4	53	62	12158	196.1
MB6	57	89	18040	202.7
MB8	56	120	20192	168.3
MUX	45	15	2646	176.4
ND/NR	56	12	4225	352.1
SDF	48	23	3853	167.5
XNR/XOR	48	14	3901	278.6
Average	52.4	35.1	6777.1	215.3

5 CONCLUSION

In this paper, we introduce ResCap, a novel approach that combines physics-guided linear models with advanced machine learning techniques to improve capacitance extraction for standard cell design. The experimental results demonstrate the superior performance of ResCap compared to traditional methods, particularly in addressing the complexities associated with 4nm technology nodes. ResCap outperforms other machine learning approaches by achieving the lowest mean absolute error, standard deviation, and outliers. In cell-level analysis, ResCap significantly reduces errors in delay and power to 0.06% and 0.16%, respectively, without any outliers, while also reducing extraction time by 215X compared to conventional field solvers. These results emphasize the potential of integrating empirical data with theoretical models to enhance predictive accuracy and computational efficiency in VLSI design.

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